

OTT Streaming Content Trend Analysis

Exploratory Data Analysis • SQL • Business Storytelling

Project report prepared as a portfolio-ready demonstration

Important data note

The included catalog is synthetic and illustrative. It is not official Netflix, Prime Video, or Hotstar data. Findings demonstrate the analytical workflow only and should not be presented as platform-specific facts.

Tools: Python, Pandas, Matplotlib, SQL, Excel

1. Executive Summary

This project demonstrates how a streaming content catalog can be audited, cleaned, explored, and translated into business hypotheses. The raw file contains 1212 unique catalog records after duplicate removal, with 12 duplicate rows removed from the original extract. The analysis covers catalog composition, genre mix, release-year distribution, country representation, content ratings, runtime, and additions by year.

The key output is not simply a collection of charts: it is a repeatable workflow, a set of SQL aggregations, and a content-strategy memo that distinguishes observed catalog patterns from hypotheses requiring viewing and cost data.

2. Business Problem and Objectives

Business problem: Content acquisition teams need to understand what their catalog contains, how the mix changes over time, and where metadata gaps prevent reliable regional or genre analysis.

Objectives: (1) inspect and clean catalog metadata; (2) quantify genre and format mix; (3) examine release and catalog-addition timelines; (4) compare countries and content ratings; (5) analyze movie runtime; (6) formulate three testable strategy actions.

3. Dataset and Scope

The raw dataset has 1224 rows before deduplication and 10 core source fields: show ID, type, title, director, country, date added, release year, rating, duration, and genre. The cleaned analysis contains 1212 unique IDs. A small number of duplicates and missing fields were intentionally introduced to make the cleaning workflow demonstrable.

Because the data is synthetic, patterns are useful for validating code and demonstrating analytical communication, not for making claims about any named service. A real deployment should use a licensed public dataset and preserve its source and license in the project.

4. Methodology

Step 1 — Profile: Inspect dimensions, column types, unique values, missingness, and duplicate show IDs.

Step 2 — Clean: Remove duplicate show IDs; parse dates and release years; replace missing categorical values with Unknown rather than silently dropping rows.

Step 3 — Engineer features: Extract numeric duration and distinguish movie minutes from TV seasons; derive year added from the date-added field.

Step 4 — Explore: Generate charts for content type, genre, release year, country, rating, movie runtime, additions by year, and genre by type.

Step 5 — Query in SQL: Provide reusable aggregations for catalog mix, genre, year, country, rating, runtime, and genre/type cross-tabs.

Step 6 — Translate to action: Write a content strategy memo with three testable actions and explicit data limitations.

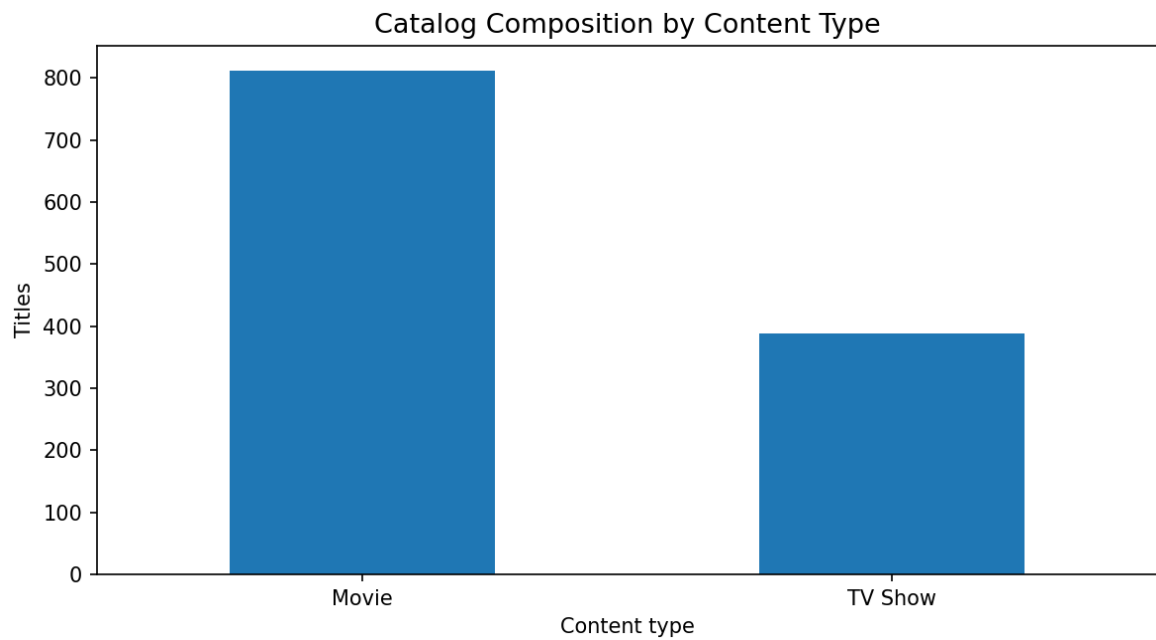
5. Data Quality and Preparation

The raw sample contains 12 duplicate rows by show ID and 223 missing cells across the source fields. Duplicate IDs are removed before aggregation. Missing director, country, rating, genre, and type values are represented as Unknown where applicable. Invalid dates are coerced to null so date-based charts do not treat malformed strings as valid dates.

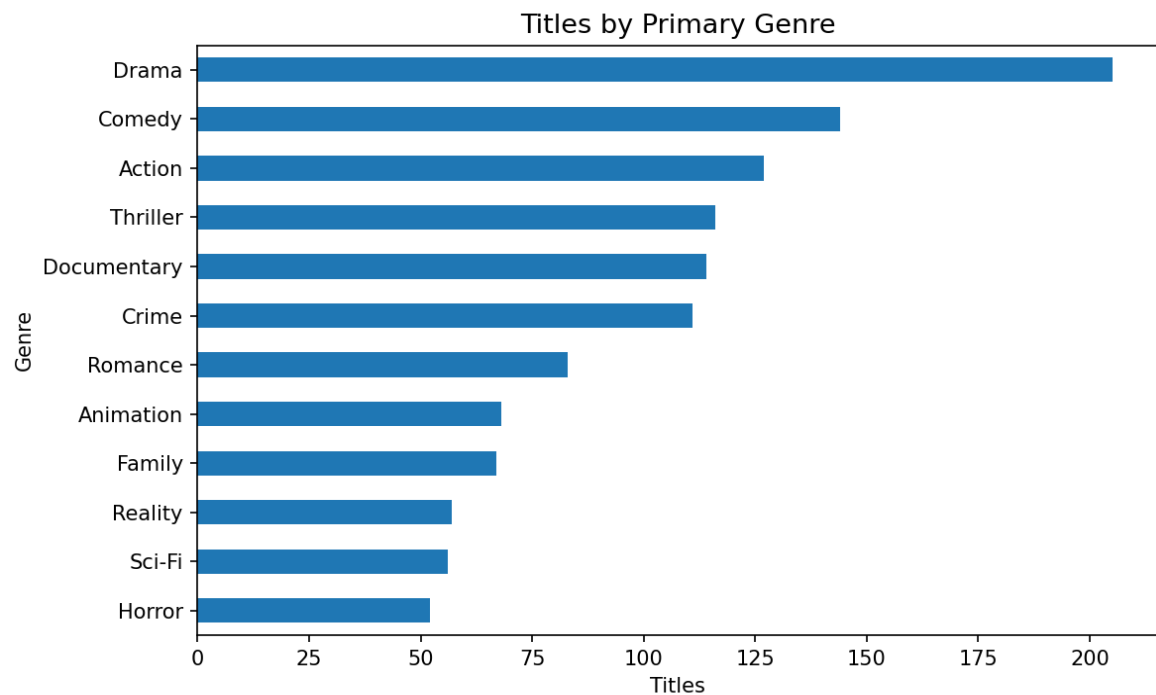
After categorical fill, remaining missing values in the main categorical fields are 0. Duration parsing extracts the first numeric token from strings such as '108 min' and '2 Seasons'; the unit is retained to avoid mixing movie minutes and TV season counts.

6. Exploratory Findings

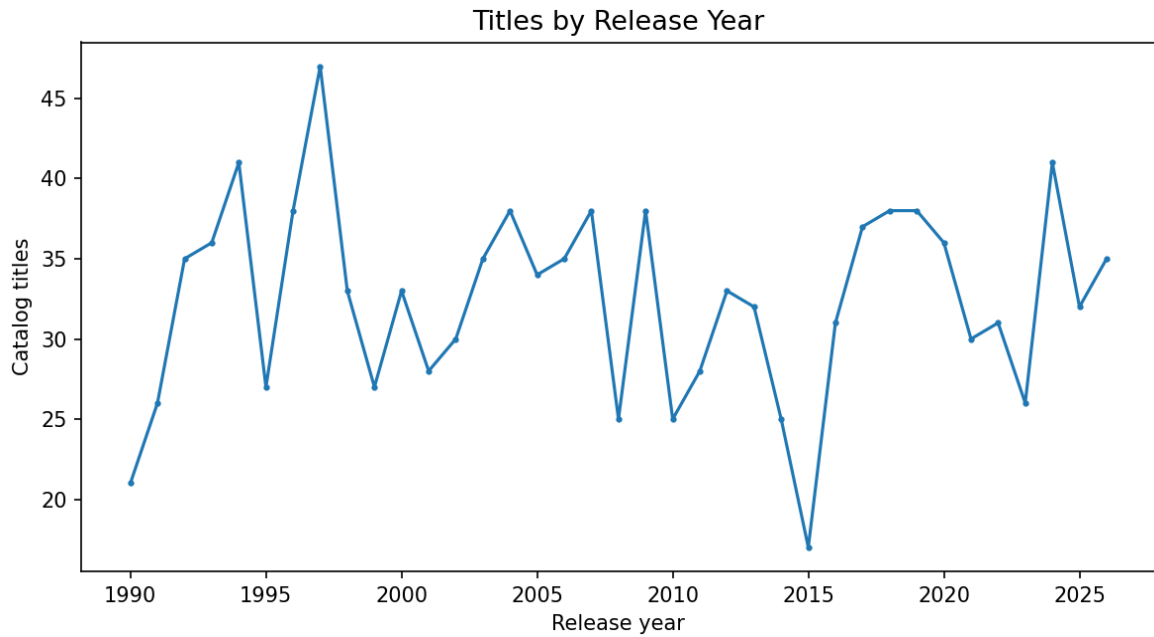
Catalog composition: Movies account for 67.7% of the cleaned sample. This is a catalog supply measure only; it does not tell us which format viewers prefer.



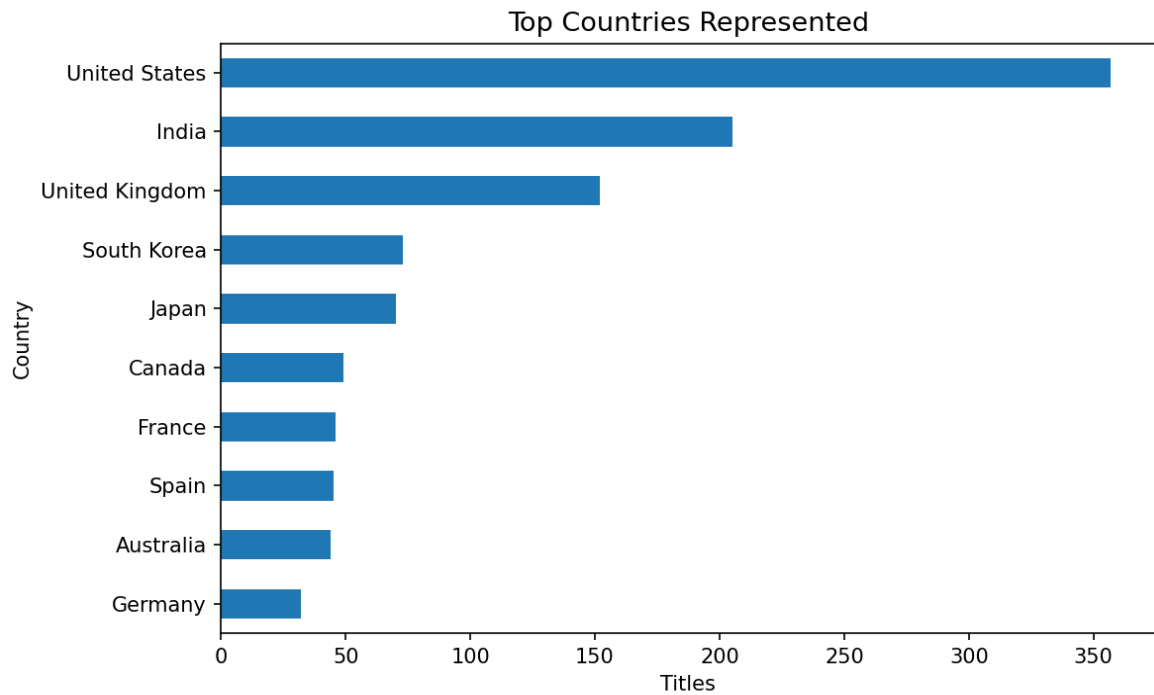
Genre distribution: Drama is the largest primary genre in this illustrative sample. Genre categories are simplified to one primary genre per title, so the result does not reflect multi-genre tagging.



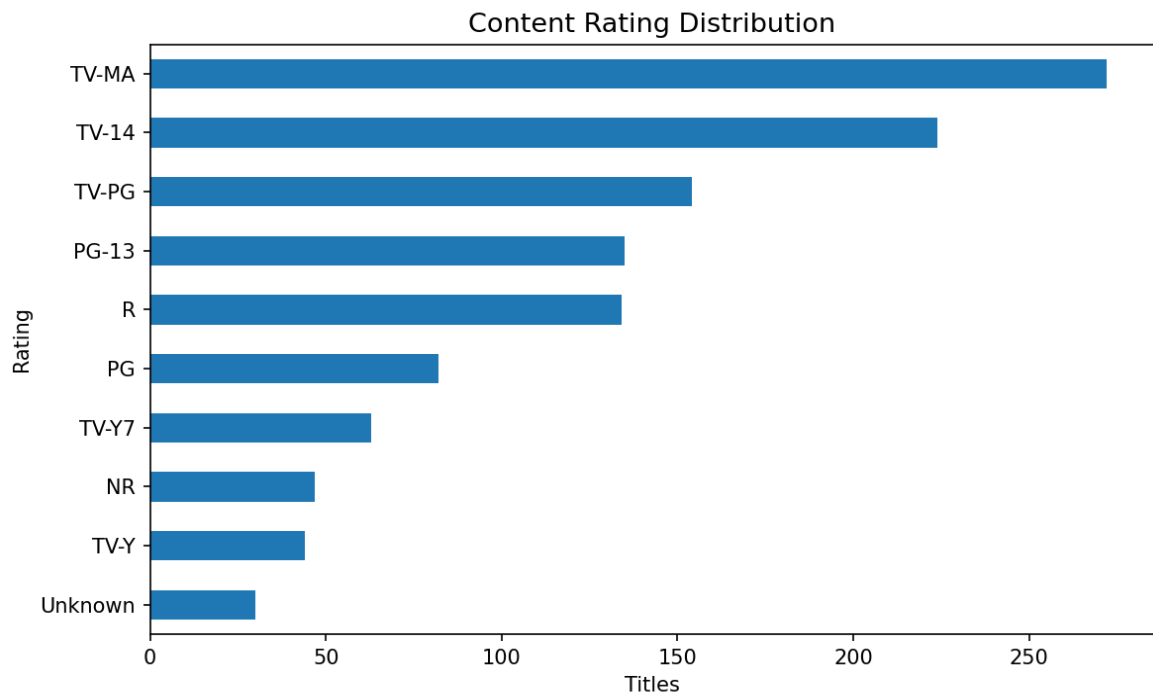
Release-year trend: This chart shows the years titles were originally released. It must not be confused with the year a title was added to a streaming catalog.



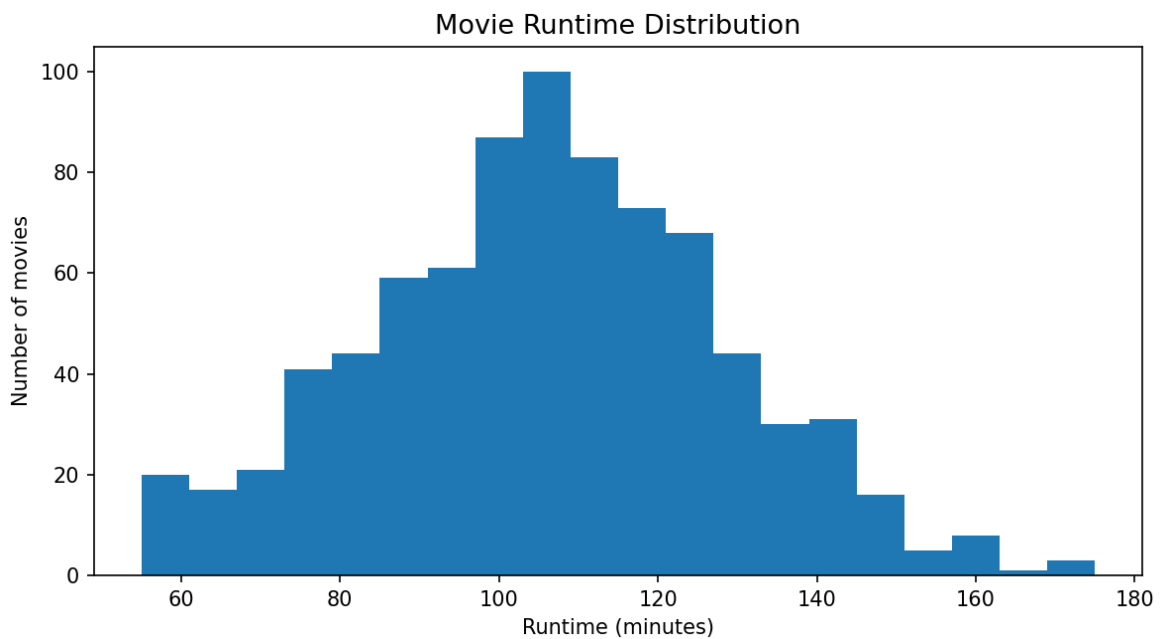
Country mix: United States is the most represented known country in the sample (357 titles). Country fields can contain multiple countries in real datasets; production-ready analysis should split multi-country strings before counting.



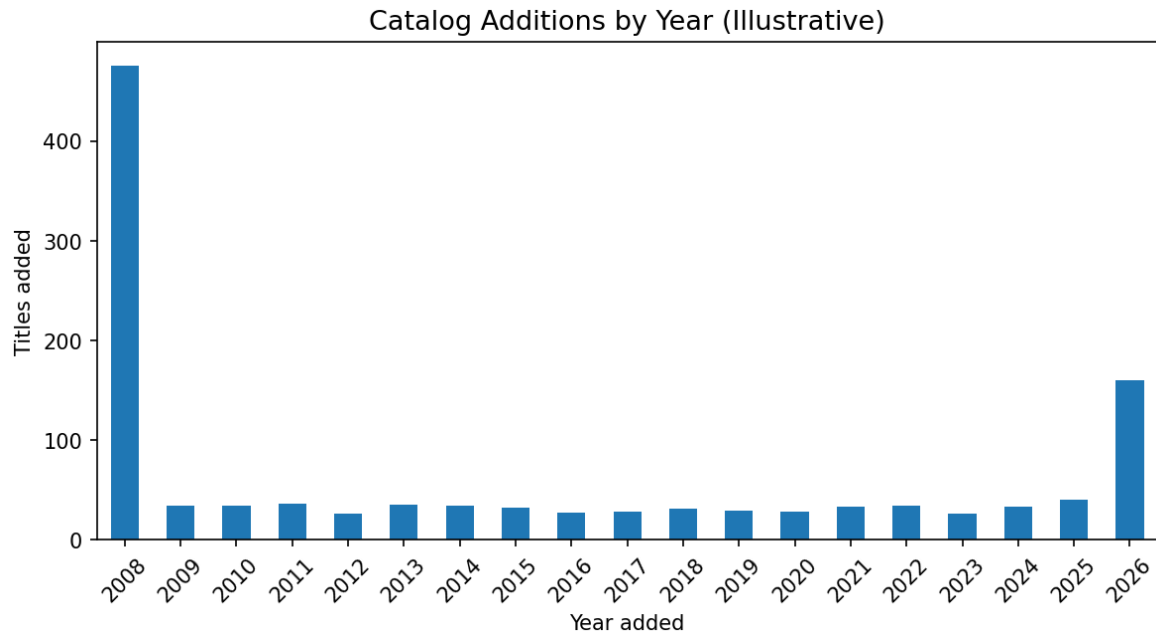
Content ratings: Ratings represent age/content classifications, not audience review scores. They can support catalog suitability analysis but cannot stand in for viewer satisfaction.



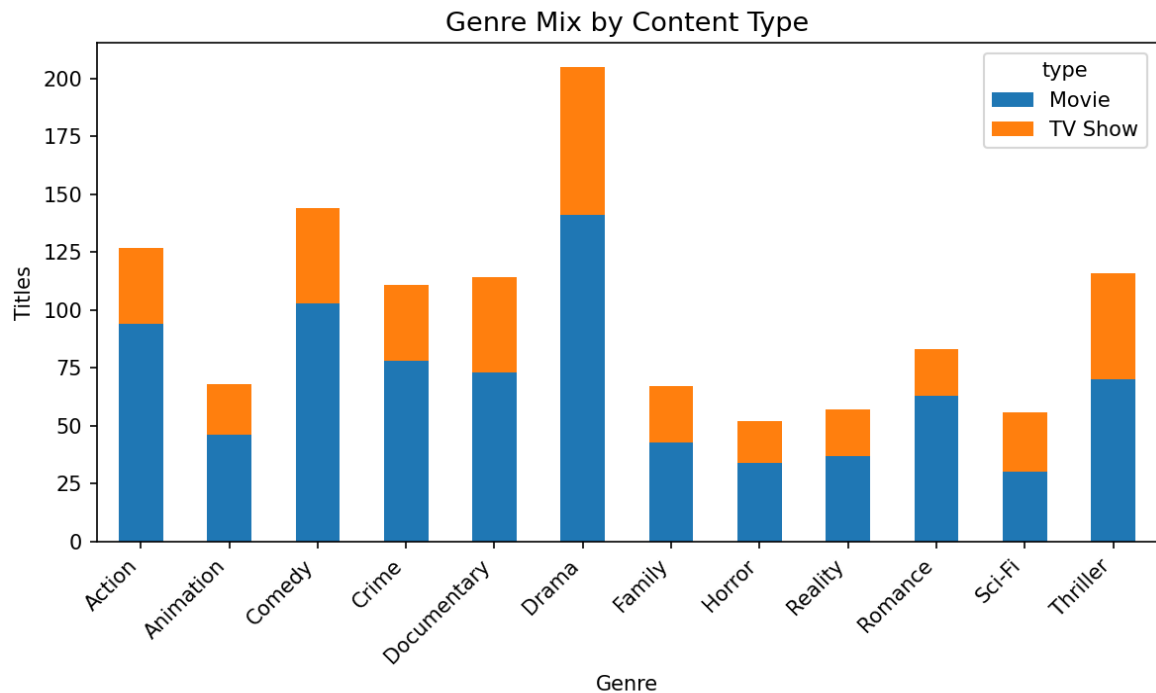
Movie runtime: Median movie runtime is 105 minutes in the illustrative sample. Runtime analysis is restricted to Movie records with minute units.



Catalog additions: The additions-by-year view is based on the date-added field, which is different from release year and is affected by missing or invalid dates.



Genre by format: The cross-tab compares movie and TV-show supply within each primary genre. It can help identify supply gaps for follow-up, but should be paired with engagement and cost metrics.



7. SQL Analysis

The ``sql/analysis_queries.sql`` file includes eight queries for catalog type mix, genre counts, release-year counts, top countries, rating distribution, movie runtime summary, genre by type, and titles added by year. Date extraction and median functions vary by SQL engine; comments identify where syntax may need adjustment.

8. Business Recommendations

Recommendation 1 — Test underrepresented genres

Choose two or three underrepresented genres for a controlled acquisition or promotion test. Compare title starts, completion rate, repeat viewing, and incremental retention with a similar control group. Do not greenlight solely because a genre has a low title count.

Recommendation 2 — Improve regional metadata

Country is Unknown for 6.0% of cleaned rows in this sample. Enrich country and language fields, then compare regional catalog coverage with local starts, completion, and subscriber growth. Separate production country from target market.

Recommendation 3 — Build a content value scorecard

Join catalog metadata to impressions, starts, watch hours, completion, retention/churn, licensing cost, release window, and acquisition terms. Rank content candidates by incremental value and strategic fit rather than catalog size alone.

9. Limitations and Risks

- All bundled catalog records are synthetic; results do not describe Netflix, Prime Video, or Hotstar.
- No views, watch hours, completion, customer cohorts, licensing cost, or retention data are present.
- Catalog representation is not equivalent to demand, popularity, profitability, or causal impact.
- One primary genre per title simplifies real-world multi-label metadata.
- Date-added values are illustrative; data quality and business definitions should be validated before use.
- SQL date extraction and median functions may require adaptation for SQLite, MySQL, SQL Server, or PostgreSQL.

10. Reproducibility and Files

Run ``pip install -r requirements.txt``, then ``python src/analyze_catalog.py``. Open ``notebooks/OTT_Content_Trend_Analysis.ipynb`` in Jupyter and execute cells top to bottom. The cleaned CSV and Excel workbook are included for immediate review. SQL queries are in ``sql/analysis_queries.sql``; the companion strategy memo is in ``reports/content_strategy_memo.md``.

11. Conclusion

This project shows the full EDA-to-business-storytelling process: validate data, clean it transparently, quantify catalog composition, visualize patterns, write SQL aggregations, and convert findings into testable next steps. The strongest next step is to replace the synthetic sample with a licensed real catalog and join it to actual engagement and cost metrics.

Prepared for portfolio and learning use. Validate all results against the dataset version you choose to publish.